SLoRa+: A Systematic Framework for Enhanced Interference Resilience in LoRaWAN

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LoRa technology has become critical in numerous IoT applications, offering long-range connections with low energy consumption. However, their low-power nature makes them vulnerable to cross-technology interference (CTI) from other wireless technologies sharing the same unlicensed frequency bands. Existing solutions address this issue through signal analysis and coding techniques. Despite these efforts, the research on CTI mitigation needs to go one step further - to design a systematic framework for enhanced interference resilience. This article proposes SLoRa+, a systematic framework that integrates symbol recovery with soft decoding to achieve synergic interference resilience for LoRa. SLoRa+'s symbol recovery employs a two-stage analysis to recover corrupted LoRa symbols at a low cost. It also estimates the confidence of the recovered symbols by considering the characteristics of both LoRa and CTI. This confidence information is then utilized in soft decoding to improve error correction capabilities. Supported by theoretical analysis and real testbed evaluations, including commercial LoRa nodes and USRP B210, our experiments under different settings, i.e., various CTI sources and frequency bands, demonstrate SLoRa+'s effectiveness. Compared with state-of-the-art techniques, SLoRa+ enhances CTI protection capability by 1.6× with only 1% of the computation cost.

CCS Concepts: • **Networks → Network protocol design;**

Additional Key Words and Phrases: LPWAN, LoRa, wireless interference, cross-technology interference (CTI)

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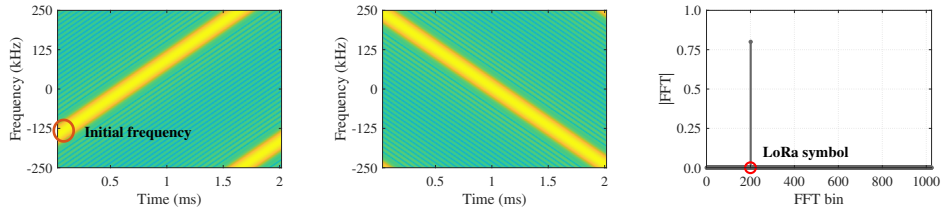
1 Introduction

With the rapid development of the **Internet of Things** (**IoT**), the communication range needs to be expanded to achieve ubiquitous connectivity. Compared with wireless technologies like Wi-Fi, Bluetooth, and ZigBee, which have restricted transmission ranges, the recently introduced **Low Power Wide Area Network** (**LPWAN**) can provide long-range connectivity at lower cost and low power consumption. These distinct attributes facilitate the rapid deployment of LPWAN in various applications. As one of the popular technologies in LPWAN, LoRaWAN has been widely used in application scenarios such as smart city [8], smart agriculture [42], environmental monitoring [48], intelligent transportation [43], and automated manufacturing [30]. According to Semtech, LoRaWAN has been adopted in 181 countries by over 166 network operators, with over 240 million end nodes and 3.2 million active gateways. And LoRa devices will account for over 50% of the deployed LPWAN devices in 2026 [4].

However, all current LoRa devices work in the unlicensed **industrial, scientific, and medical** (**ISM**) bands. The choice of the unlicensed spectrum leads to **Cross-Technology Interference** (**CTI**), which can negatively impact the network's performance. According to [23, 27, 40, 47], due to the inherent characteristics of LoRa such as long communication distance, low energy consumption, and long packet duration, the packet rate of LoRa devices may drop below 50% when affected by CTI. This large amount of packet loss subsequently leads to a significant increase in transmission delay and energy consumption.

To address this fundamental CTI issue, researchers have proposed multiple lines of solutions. Techniques like PSR [47] directly analyze the received signal under CTI to recover corrupted symbols. Although effective, it fails to recognize the reliability difference across recovered symbols - its recovery reliability decreases when a LoRa symbol is corrupted by longer CTI, while its heavy computation overhead is also unfriendly to low-power end devices. **Forward Error Correction** (**FEC**) techniques, such as [11, 36], design customized coding schemes to shield LoRa from CTI. Since these methods have zero access to physical layer information, their reliability is directly constrained by the chosen coding method and coding rate. Until now, a systematic approach for enhanced interference resilience under CTI is still absent.

To fill this gap, we propose SLoRa+, a systematic framework that integrates symbol recovery with soft decoding. For a corrupted symbol, we adopt a two-step signal analysis strategy from coarse to fine-grained granularity in the amplitude and frequency domain. This approach effectively avoids the need for complex time-frequency analysis on long-duration LoRa symbols. Firstly, our amplitude analysis carefully examines the statistical characteristics of LoRa and CTI to identify signal segments that are potentially corrupted by CTI. Subsequently, the frequency analysis searches for LoRa's unique features in the frequency domain for CTI verification and symbol recovery. Through these lightweight procedures, SLoRa+ can recover corrupted symbols with high reliability and at a considerably lower cost compared with the state-of-the-art method PSR [47]. For recovered symbols, SLoRa+ estimates their reliability confidence, considering relevant statuses of CTI and LoRa, such as the CTI duration with a LoRa symbol, **signal-to-noise ratio** (**SNR**), and **spreading factors** (**SF**). Specifically, SLoRa+ establishes a theoretical mathematical model in the **additive white Gaussian noise** (**AWGN**) channel for symbol confidence estimation and validates it with simulation. Furthermore, it extends this theoretical model to more realistic conditions and



(a) LoRa signal (b) A standard downchirp (c) FFT spectrum

Fig. 1. LoRa demodulation including dechirping and FFT. (a) shows a LoRa chirp with the shifted initial frequency f_s; (b) shows a standard downchirp signal; (c) The FFT spectrum of the LoRa signal after dechirping.

extracts distinct bit error patterns within a LoRa symbol to accurately measure bit confidence. This estimated bit confidence is used as the input in SLoRa+'s soft decoding to improve interference resilience. By doing so, SLoRa+'s systematic framework ensures synergic interference resilience under CTI while maintaining cost-effectiveness.

Specifically, SLoRa+ has the following contributions:

(1) To the best of our knowledge, this is the first systematic framework designed to achieve synergic interference resilience for LoRa under CTI .
(2) The architecture of SLoRa+ includes lightweight symbol recovery and soft decoding. For symbol recovery, SLoRa+ adopts a two-step strategy of amplitude and frequency analysis. Specifically, the amplitude analysis of SLoRa+ has minimal computation, followed by frequency analysis to enhance reliability by carefully examining LoRa's unique frequency domain patterns. After recovering corrupted symbols, SLoRa+ accesses their confidence by considering various factors and further leverages this estimation in soft decoding to augment the protection capability. As a result, SLoRa+ enables synergic reliability for low-power LoRa communication against CTI in a lightweight way.
(3) We evaluate our design with theoretical analysis and experiments via real testbeds. The comprehensive results illustrate that our solution can provide reliable protection against CTI. It achieves 1.6x interference resilience and operates 100x faster than the existing state-of-the-art technique.

2 Background and Motivation

In this section, we first introduce the LoRa technology in the physical layer, which includes the modulation/demodulation method and coding/decoding scheme. Then, we analyze the CTI impacts for LoRa transmissions in the unlicensed ISM band.

2.1 LoRa PHY

LoRa modulation and demodulation. LoRa adopts **Chirp Spread Spectrum** (**CSS**) to modulate data [7]. It uses a series of upchirp signals with different initial frequencies to represent LoRa symbols. The LoRa signal can be expressed as

$$x_s(t) = A_s e^{j2\pi\left(-\frac{BW}{2}+\frac{1}{2}kt+f_s\right)t}, \tag{1}$$

where A_s is the signal amplitude of the transmitted signal and k = $\frac{BW^2}{2^{SF}}$ is the frequency change rate, which is determined by the **spread factor** (**SF**) and the **bandwidth** (**BW**). A basic unchirp sweeps the transmission bandwidth from $-\frac{BW}{2}$ to $\frac{BW}{2}$. CSS modulates the transmitted symbol s by changing the initial frequency as shown in Figure 1(a).

The demodulation stage includes the dechirping and **Fast Fourier Transform** (**FFT**) operation. Specifically, dechirping requires a dot multiplication of the received signal with a standard downchirp whose frequency decreases linearly with time, which can be expressed as $x_0^{-1}(t) = e^{-j2\pi(-\frac{BW}{2}+\frac{1}{2}kt)t}$. Figure 1(b) shows a standard downchirp. After dechirping, the signal contains only the frequency component f_s. The LoRa symbol s is then determined by identifying the frequency bin in the FFT spectrum that has the maximum amplitude, as depicted in Figure 1(c). This frequency bin directly relates to the original data symbol encoded in the chirp's initial frequency.

LoRa coding and decoding. LoRa physical layer adopts FEC with Hamming codes that support four coding rates CR $\in \{4/5, 4/6, 4/7, 4/8\}$ [67]. The length of the codeword after encoding is denoted as m and $m \in \{5, 6, 7, 8\}$. When m is 5 or 6, it can identify errors, but it lacks correction abilities. When m is 7 and 8, a one-bit error within a codeword could be corrected. In addition, LoRa adopts a diagonal interleaver to interleave the bits within each block, where the LoRa block length is $SF \times m$ bits, instead of the packet length. The receiver performs complementary deinterleaving and decoding procedures to obtain the transmitted data.

2.2 LoRa in ISM Band

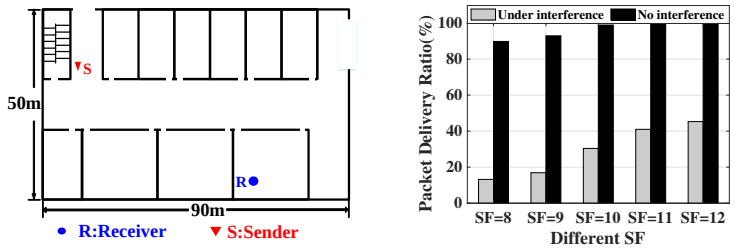
To enable long-range communication for low-power devices, Semtech introduces the LoRa. It utilizes CSS modulation, which enhances resilience against environmental noise. To reduce deployment costs, LoRa operates in the unlicensed Industrial, Scientific, and Medical (ISM) band. Based on the ISM frequency band, LoRa works in the sub-1 GHz and 2.4 GHz bands. However, the sub-1 GHz ISM band has regional regulations. For example, LoRa operates at 915 MHz in North America, 868 MHz in Europe, and 470 MHz in China. These regional differences lead to extra hardware complexity for LoRa devices. To simplify global deployment, Semtech has introduced LoRa chips (e.g., SX1280, SX1281 [5]) which operate at 2.4 GHz, a globally available frequency without regional limitations. The 2.4 GHz LoRa chip uses the same modulation and coding method as the sub-1 GHz LoRa chip. Compared with traditional IoT technologies such as Wi-Fi, Bluetooth, and ZigBee, it provides long-range communication based on CSS modulation [5].

2.3 CTI Impacts

Since LoRa operates in the unlicensed ISM band, it inevitably needs to compete with other wireless technologies for limited spectrum resources, resulting in CTI, i.e., low-power LoRa communication is susceptible to interference from other coexisting wireless technologies in the same frequency band. For example, in the 915 MHz ISM band, LoRa could be affected by the Wi-Fi Halow [51] (i.e., IEEE 802.11ah) technology. As for the worldwide 2.4 GHz band, it is crowded with various popular wireless technologies such as Wi-Fi, ZigBee, and Bluetooth, and LoRa devices operating in this band also have the serious CTI problem.

Given the long-range communication characteristics of LoRaWAN, the communication range can reach 10 km, and the LoRa signal power at the receiver is extremely low, e.g., -132 dBm [5]. This ultra-low-power communication could be corrupted by ongoing CTI with high power. Additionally, LoRa transmissions typically have a long duration, e.g., a single packet could last 400 ms [5], while a normal inter-packet duration in Wi-Fi is less than 3 ms [47]. This extended on-air time further increases the likelihood of LoRa corruption, especially in the crowded 2.4 GHz ISM band.

Although the interference from multiple LoRa transmissions has been extensively studied [12, 13, 15, 25, 29, 35, 44, 53, 57–59, 63, 68, 71, 72], these existing solutions are not applicable to address CTI, because of the inherent difference between the LoRa signal and CTI. For example, multiple LoRa transmissions can be distinguished by leveraging the unique characteristics of LoRa signals in the time and frequency domains. In contrast, CTI stems from different wireless technologies, which exhibit diverse and unpredictable interference patterns. These patterns differ significantly



(a) Indoor experiment scenario (b) Packet delivery ratio comparison

Fig. 2. Impact of cross-technology interference (CTI) on LoRa communication.

from those caused by collisions among LoRa packets and requires a dedicated approach to handle CTI.

To verify the effects of CTI on LoRa transmissions, we have conducted the following preliminary experiments in a teaching building. We use the 2.4 GHz SX1280 LoRa node to transmit 1000 packets at different SF settings. SF determines the duration of LoRa symbols and the ability to resist noise. The transmission power is set to 12.5 dBm. Figure 2(a) presents the scenario map. Due to the complex indoor environment and numerous obstructions, the SNR is approximately -10 dB. We conduct two sets of experiments in the middle of the night and in the morning, respectively, to test the LoRa **packet delivery rate** (**PDR**). In the middle of the night, no Wi-Fi devices are connected to the Wi-Fi AP in the building, so there is almost no CTI in the environment. In the morning, there are more than 5 Wi-Fi devices connected to the same Wi-Fi AP, so the interference is strong. Figure 2(b) illustrates that standard LoRa performs well in environments without interference. It successfully decodes 89.8% of packets when SF = 8 and nearly 100% of packets when SF reaches 10. In contrast, the packet delivery ratio is only 13.2% under CTI when SF is 8. Even though an increase in SF enhances resilience, the packet delivery ratio is less than 50%, significantly lower than in cases without interference. These experiments highlight the serious impact of CTI on the reliability of LoRa communications.

3 System Overview

3.1 Overview

SLoRa+ is a software solution focused on solving the impact of CTI on LoRa communication. Our design is implemented on a LoRa receiver and is compatible with all existing LoRa transmitters. SLoRa+ includes two primary components: Lightweight Symbol Recovery (Section 4) and Soft Decoding (Section 5). Figure 3 demonstrates the system overview of SLoRa+.

Lightweight symbol recovery. The objective of SLoRa+ is to directly analyze the corrupted LoRa signal from coarse to fine granularity in a cost-effective manner for symbol recovery. Specifically, it initially examines the amplitude changes with minimal computational overhead in the time domain, and then further employs the frequency analysis to carefully examine the unique signal characteristics of LoRa in the frequency domain. By conducting a hierarchical search of the signal, SLoRa+ avoids the complex time-frequency analysis like Short-time Fourier Transform (STFT) which is time-consuming for the long-duration LoRa symbols and manages to provide reliable symbol recovery at a low cost.

Soft decoding. To enhance LoRa communication resilience against CTI, SLoRa+ adopts soft-decision decoding with probabilistic demodulation outputs. Specifically, SLoRa+ exploits the unique features of the LoRa signal that are inherently different from CTI for symbol confidence estimation and further leverages the distinct bit error patterns within a LoRa symbol for the accurate estimation

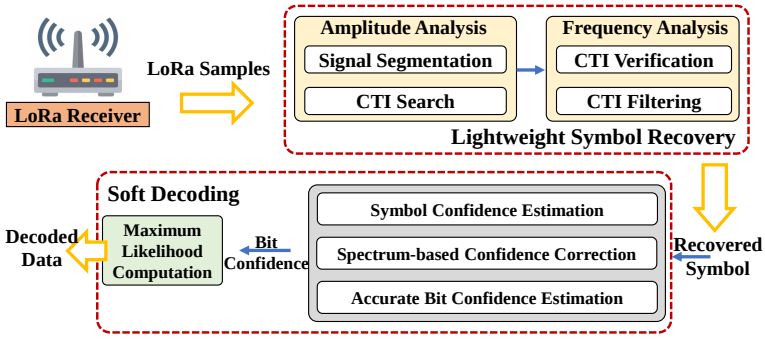


Fig. 3. System overview of SLoRa+.

of the bit confidence. Subsequently, SLoRa+ leverages the bit confidence in its soft decoding for enhanced error correction. Note that even though SLoRa+ focuses on Hamming code as Hamming code is adopted in standard LoRa encoding schemes [67], our basic idea can be applied to other coding techniques.

3.2 Challenges

To provide a CTI-resilient framework that is both cost-effective and reliable, we need to overcome several critical challenges:

1. Universal capability. Given that LoRa coexists with other wireless technologies in the unlicensed ISM band, LoRa transmissions may interfere with signals from various other wireless protocols such as Wi-Fi, Bluetooth, ZigBee, and more. This necessitates the design of a unified protection framework that effectively guards against diverse CTI scenarios without prior knowledge of the interference's characteristics, such as its type, intensity, or duration.

2. Reliable protection. To mitigate CTI, existing research either analyzes signals at the physical layer or introduces additional redundancy at higher layers through coding techniques. In contrast, we need to design a systematic framework to enhance the protection capability against CTI.

3. Cost efficiency. As our focus is on low-power communication and applications, the method needs to provide reliable interference protection at a low computational and storage cost.

4 Lightweight Symbol Recovery

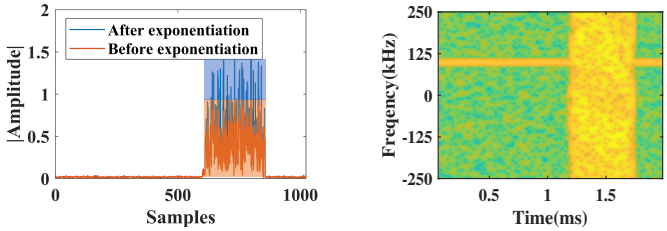
This section introduces our lightweight symbol recovery technique. SLoRa+ first adopts the amplitude analysis with minimal computational cost to locate suspicious signal parts that could be impacted by CTI. Then it further analyzes them in the frequency domain for verification and recovers corrupted LoRa symbols leveraging the partially clean LoRa signal.

4.1 Amplitude Analysis

SLoRa+'s amplitude analysis includes signal segmentation and CTI search. We focus on the scenario where LoRa signal suffers from high power CTI, which is dominant due to the long-range and low-power characteristics of LoRa.

4.1.1 Signal Segmentation. For a LoRa chirp distorted by CTI, its sampled received signal at the LoRa receiver is denoted as $y_s[n]$.

$$y_s[n] = \begin{cases} hx_s[n] + w[n] & n \in \mathcal{N}_1 \\ hx_s[n] + w[n] + I[n] & n \in \mathcal{N}_2 \end{cases}. \tag{2}$$



(a) Signal preprocessing (b) Spectrogarm of Fig.(a)

Fig. 4. Comparison of clean LoRa signal parts and the signal parts affected by CTI. (a) Time-domain waveform showing the effect of CTI and the amplified amplitude difference after signal preprocessing; (b) corresponding spectrogram of (a), showing the LoRa signal under CTI in the time–frequency domain.

In Equation (2), $\mathcal{N}_1$ and $\mathcal{N}_2$ represent the set of clean LoRa signal segments and CTI-contaminated signal segments, respectively. $w[n]$ represents the complex Gaussian white noise in the environment. Specifically, $w[n] \sim \mathcal{CN}\left(0, \sigma^2\right)$, where $\sigma^2 = \frac{N_0}{2}$ and N_0 is the signal-sided noise power spectral density. $I[n]$ is the CTIs that could result from multiple wireless technologies/transmissions. Note that those $I[n]$ that can negatively affect LoRa demodulation basically have higher energy than the received LoRa signal. And h represents the channel response and is usually constant over short periods. We address the variation of h under dynamic environmental conditions in Section 4.2.

According to the standard LoRa demodulation in Section 2.1, after dechirping, the clean LoRa signal becomes $\widehat{y_s[n]} = hA_s e^{j2\pi f_s \frac{n}{N}} + \widehat{w[n]}$, where N is the number of signal samples within one LoRa chirp and A_s the signal amplitude of the transmitted signal. Sepcifically, the noise part after dechirping $\widehat{w[n]}$ still accord with complex Gaussian white noise distribution, and its amplitude envelope $|\widehat{w[n]}|$ can be expressed as a random variable with Rayleigh distribution. Then, the amplitude of clean LoRa signal after dechirping $A = |\widehat{y_s[n]}|$ can be regarded as the sum of the original signal and the multipath noise obeying the Rayleigh distribution, which can be approximated with Rician distribution $f\left(A \mid h\bar{A}_s, \sigma\right)$, where the $h\bar{A}_s$ is the mean amplitude of the received signal and σ is the standard deviation of the noise. In contrast, CTI, due to its diverse communication protocols, shows varying amplitude patterns. This motivates us to distinguish clean LoRa signal from CTI, without prior knowledge of CTI.

Based on the above theoretical analysis, it is feasible to employ the changes in amplitude as a detection metric for CTI. Specifically, we get the amplitude of the signal after dechirping

$$A[n] = |y_s[n] \cdot x_0^{-1}[n]|, \tag{3}$$

where $A[n]$ denotes the amplitude of the signal after dechirping and $x_0^{-1}[n]$ is the standard downchirp. Further, to mitigate interference from environmental noises and enhance reliability, our methodology integrates signal preprocessing and a window sliding algorithm to detect changes in amplitude.

• **Signal preprocessing.** Firstly, to amplify the amplitude difference between pure LoRa signal parts and CTI parts, SLoRa+ performs normalization and exponentiation before starting the analysis.

$$A'[n] = e^{\frac{A[n]-\min(A[n])}{\max(A[n])-\min(A[n])}}. \tag{4}$$

The exponential function increases monotonically and the slope gradually increases, which can heighten the contrast ratio of the amplitude value. As shown in Figure 4(a), it can be seen that after the signal preprocessing of $A[n]$, the CTI part is nearly doubled, and the part of the clean LoRa

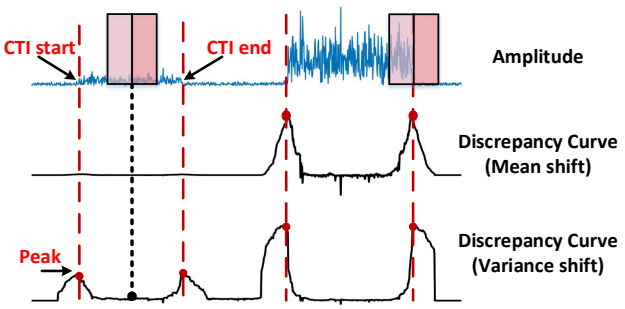


Fig. 5. Window sliding algorithm for detecting pure LoRa and CTI segments.

signal becomes smaller, which shows that it can increase the gap between clean signal parts and CTI parts so that the model can better identify interference.

• **Window sliding.** The main idea of window sliding is to calculate a discrepancy index as a quantitative measure of deviation between intercepted signals during the first half of the window win_1 and the second half of the window win_2. The discrepancy index is denoted as

$$d\left(A'_{a:t}, A'_{t:b}\right) = c\left(A'_{a:b}\right) - \left[c(A'_{a:t}) + c\left(A'_{t:b}\right)\right]. \tag{5}$$

In Equation (5), $A'_{a:b}$ means the amplitude of the segment with the starting index a and the end index b. The two sub-windows are $win_1 = A'_{a:t}$ and $win_2 = A'_{t:b}$, respectively. t is the midpoint of the window which is denoted as $t = a + \frac{b-a}{2}$. $c(\cdot)$ is a cost function that measures the degree to which samples within a window deviate from the empirical observations of the window. The window size is empirically set to $\frac{2^{SF}}{8}$ in our design.

Because the amplitude distributions of the clean LoRa signal and the CTI are different, statistical features can be used to distinguish those signal parts. We utilize the mean shift and variance shift to calculate the cost function simultaneously. Specifically,

$$c_M\left(A'_{a:b}\right) = \sum_{n=a+1}^{b} \left|A'_n - \bar{y}_{a:b}\right|^2, \tag{6}$$

$$c_V\left(A'_{a:b}\right) = (b-a)\log\sigma^2_{a:b}, \tag{7}$$

where $\bar{y}_{a:b}$ and $\sigma^2_{a:b}$ are the empirical mean and variance of the segment $A'_{a:b}$. Using the mean and variance at the same time can prevent the model from being unable to divide the signal when a certain type of statistical feature is not obvious.

Our previous analyses indicate that the amplitude follows the Rice distribution when there is no CTI in the window. On the other hand, CTI causes a sharp rise in the discrepancy index (Equation (5)). The maximum points are the start and the end of CTI. This is because at the start and the end of CTI, the win_1 and win_2 cover signals with dramatically different distributions, resulting in the highest discrepancy. It should be emphasized that our focus is on distribution differences rather than specific CTI distribution, given the insufficient prior knowledge about CTI. Figure 5 shows the received LoRa signal. The discrepancy indexes reach the highest at the CTI boundary, which could be exploited to detect suspicious signal segments that might contain CTI. Moreover, reliance solely on Mean shift may overlook low-power CTI, but employing variance and mean simultaneously enhances the robustness of signal segmentation.

Algorithm 1 is the computation procedure for signal segmentation. Given the sliding window, whose midpoint is t, we initially compute the discrepancy index $Z[t]$. Then we detect change points

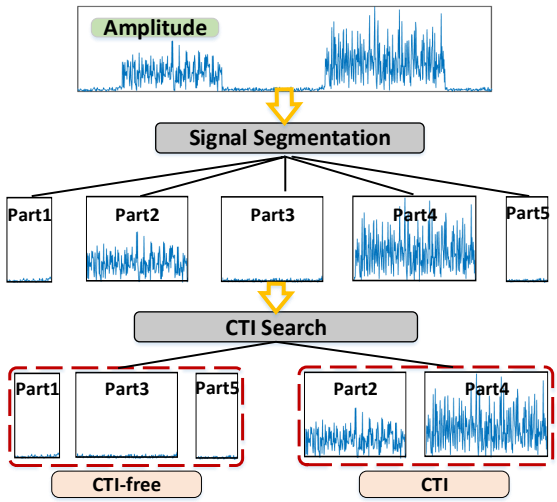


Fig. 6. The amplitude analysis example of SLoRa+.

ALGORITHM 1: Window Sliding

Input: The amplitude after signal processing A'; Cost function $c(\cdot)$; Half-window width ω; Peak search procedure $findpeaks(\cdot)$.

Initialize Z $\leftarrow$ zeros(1,2^{SF});
for $t = \omega, 2\omega \ldots 2^{SF} - \omega$ **do**
  $p \leftarrow t - \omega : t$;
  $q \leftarrow t : t + \omega$;
  $r \leftarrow t - \omega : t + \omega$;
  $Z[t] \leftarrow c(A'_r) - [c(A'_p) + c(A'_q)]$;
end
$L \leftarrow findpeaks(Z)$;

Output: L, the change point indexes.

(local maximum points) via the peak analysis function provided by the Signal Processing Toolbox of MATLAB (R2022a). Figure 6 shows an example of signal segmentation. Based on the discrepancy computation results, the entire signal is divided into 5 segments in this example for further analysis.

In summary, the window sliding algorithm emphasizes the variance in amplitude distribution between pure LoRa and CTIs and does not need any prior knowledge about CTI, i.e., type, power, or duration. At the same time, SLoRa+ employs various statistical features (mean shift and variance shift) to detect CTI, enhancing the model's precision and robustness, which can be used in various complex situations.

4.1.2 CTI Search. After dividing the received LoRa signal into multiple segments, we further search for CTI via the following procedures. The basic idea is to examine statistical values for each segment and subsequently assign these segments into one of two categories: CTI or CTI-free.

Specifically, we first estimate the mean amplitude for every segment. Based on these computed means, the segment exhibiting the minimum power is divided into the CTI-free set, while the segment with the maximum is added to the CTI set. We then employ the nearest neighbor principle, i.e., iteratively selecting a segment and calculating its distance to both the CTI and CTI-free sets and

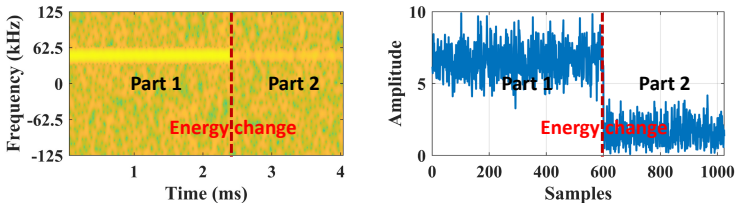


Fig. 7. Example of symbol energy variation in one LoRa chirp signal without CTI.

deciding which category it belongs to according to the distance. Note that we aim at ensuring that the segments in the CTI-free set are indeed real LoRa signals, and we can tolerate the misclassification of a LoRa segment into the CTI set. This is because Section 4.2 further reliably examines LoRa frequency characteristics to verify the classification performance. As a result, we introduce an additional penalty factor α when computing the distance to the CTI-free set, and the penalty is calculated based on the average of the minimum and maximum values.

4.2 Frequency Analysis

With the signal segments identified by amplitude analysis, our frequency analysis examines LoRa features for further verification, which offers better reliability at a larger cost.

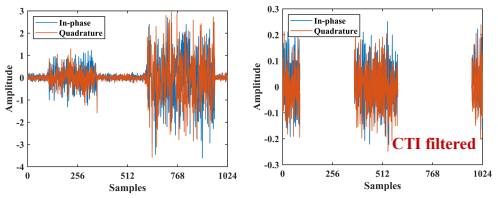
4.2.1 CTI Verification. Amplitude analysis cuts the received signal into segments and searches for segments that may be corrupted by CTI. The number of segments is constantly changing, depending on the number of CTI parts that the symbol is disturbed by. However, due to the long LoRa symbol duration and potential dynamic environment, the energy of the LoRa signal may significantly fluctuate in practice, potentially leading to classification inaccuracies. Figure 7 shows the received LoRa chirp without CTI. It can be seen that the signal energy drops significantly due to environmental variation. Since SLoRa+ checks amplitude variations in Section 4.1, it might erroneously categorize a CTI-free segment as a CTI segment.

To provide more reliable CTI detection, SLoRa+ further examines the LoRa signal characteristics in the frequency domain, which is completely different from other types of interference. Specifically, we define the Single Frequency Factor Coefficient (SFFC).

$$\text{SFFC} = \frac{(\max(R[n]))^2}{\sum_{n=1}^{N} R[n]^2}, \tag{8}$$

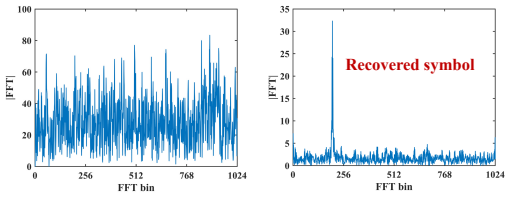
where $R[n] = |\text{FFT}(y_s[n] \cdot x_0^{-1}[n])|$ and represents the amplitude of the FFT spectrum for the LoRa signal after dechirping. Since the CSS modulation technique, as mentioned in Section 2.1, the LoRa signal after dechirping has a dominant frequency component in the frequency spectrum, i.e., this frequency component has a much larger value than all other components and is recognized by the LoRa demodulation result. In this case, the SFFC result is high. In contrast, since other wireless communication technologies are not based on CSS, the segments corrupted by CTI lack the dominant frequency after the FFT operation, resulting in significant changes in SFFC. Motivated by this, we calculate the SFFC for each segment identified as CTI in Section 4.1. Segments marked as CTI and have a high SFFC (i.e., SFFC > 0.2) are reclassified as CTI-free. It should be noted that the SFFC also depends on the segment length and SNR. For a CTI-free segment, the SFFC increases with the segment length and SNR.

Since we initially rely on the amplitude analysis in Section 4.1 to select segments potentially affected by CTI, our SFFC examination only applies to these segments in the CTI set to reduce the probability of misclassification, which avoids a comprehensive traversal through the entire long



(a) LoRa signal corrupted by CTI (b) Amplitude after CTI filtering

Fig. 8. Amplitude before and after SLoRa+.



(a) FFT result before recovery (b) FFT result after recovery

Fig. 9. FFT before and after SLoRa+'s recovery.

LoRa symbol for time-frequency analysis. In practice, some signal segments could be affected by CTI but classified as CTI-free sets. This does not impact SLoRa+, as this CTI's energy is typically minimal and could be mitigated by standard LoRa demodulation. The two-stage analysis offers good reliability under various scenarios, as validated in Section 7.

4.2.2 CTI Filtering. After identifying LoRa signal segments corrupted by CTI, we filter out the CTI and utilize the remaining clean LoRa signal to demodulate the symbol. Similar to the traditional demodulation method, we discard the CTI parts by setting these parts to 0. Then we follow the standard LoRa demodulation procedures, which include dechirping and FFT for the symbol recovery.

Figure 8(a) shows an example of a corrupted LoRa signal. SLoRa+ first identifies and filters the CTI parts to get the pure LoRa signals as shown in Figure 8(b). Figure 9 shows the corresponding demodulation results of Figure 8. The FFT result of the original signal before recovery lacks a dominating frequency component, meaning the demodulation fails. When SLoRa+ filters out CTI, the FFT result demonstrates a high peak, which is exactly the transmitted LoRa symbol. SLoRa+ successfully recovers the corrupted LoRa symbol via its amplitude and frequency analysis.

5 Soft Decoding

Although SLoRa+ analyzes the corrupted signal for symbol recovery, its performance is unavoidably influenced by the CTI ratio within a LoRa symbol. SLoRa+ cannot recover symbols that have been corrupted by CTI with a long duration (see Sections 6 and 7.1). Consequently, SLoRa+ employs soft decoding to further enhance communication resilience under CTI. It first examines LoRa and CTI characteristics for confidence estimation of recovered symbols and their bits, followed by maximum likelihood computation.

Given that standard LoRa packets already implement hamming coding [50], SLoRa+ directly utilizes this opportunity in its soft decoding without designing its own decoding schemes. This ensures the compatibility of SLoRa+ with existing LoRa packet formats and applications. It should be noted that the general design philosophy of SLoRa+ could be directly applied to other custom coding solutions by simply incorporating the estimated symbol confidence with the corresponding decoding procedures.

5.1 Symbol Confidence Estimation in AWGN

According to the CSS demodulation, the frequency bin exhibiting the highest amplitude will be selected as the symbol's value after performing dechirping and FFT, which is due to the orthogonality of the LoRa signal [54]. Specifically, with the received signal $y_s[n]$ that corresponds to a transmitted signal $x_s[n]$ over an AWGN channel with CTI, the correlator output of the LoRa demodulator is

$$\sum_{n=0}^{2^{SF}-1} y_s[n] \cdot x_i^{-1}[n] = \begin{cases} 2^{SF}(hA_i + \varphi_i + CTI_i) & i = s \\ 2^{SF}(\varphi_i + CTI_i) & i \neq s \end{cases}, \tag{9}$$

where h and A are the channel gain and amplitude during the transmission of symbol s. φ_i is the complex Gaussian noise process and CTI_i stands for the CTI.

Intuitively, demodulation accuracy is related to channel quality, particularly the power of noise and the degree of CTI. The higher the energy of a compensated frequency peak, the higher the chance that the frequency peak carries the transmitted symbol. Based on this understanding, we define $\delta_i = |\varphi_i + CTI_i|$, which denotes the magnitude of the envelope of noise and CTI. Therefore, the probability of a LoRa symbol error can be expressed as follows

$$P_{e|s} = \Pr\left[\max_{i,i\neq s}(\delta_i) > |hA_s + \varphi_s + CTI_s|\right]. \tag{10}$$

Figure 9(a) presents a case of a corrupted LoRa signal, where CTI power significantly exceeds LoRa, greatly increasing the probability of error. However, SLoRa+'s lightweight symbol recovery can accurately identify and filter out CTI. At the same time, it can also calculate the $\text{CTI}_{\text{ratio}}$ for each LoRa symbol. Consequently, the pure LoRa signal parts obtained by SLoRa+ are approximated to signals transmitted in an **AWGN** channel, and the probability of symbol error becomes

$$P'_{e|s} = \Pr\left[\max_{i,i\neq s}(\delta'_i) > |hA_s + \varphi_s|\right]. \tag{11}$$

Furthermore, $\delta'_i = |\varphi_i|$ is a random variable that satisfies the Rayleigh distribution, and its **cumulative distribution function** (**CDF**) is $F_{\delta'_i}(\delta') = 1 - \exp\left[-\frac{\delta'^2}{2\sigma^2}\right]$. Let $\rho = \max_{i,i\neq s}(\delta'_i)$ depicts a random variable for the maximum of $2^{SF} - 1$ i.i.d. Rayleigh random variables[16]. The CDF for ρ can be given as

$$F_\rho(x) = \left[1 - \exp\left[-\frac{x^2}{2\sigma^2}\right]\right]^{2^{SF}-1}. \tag{12}$$

Given Equation (11) and 12, the amplitude $|hA_s + \varphi_s|$ of the pure LoRa signal parts conforms to the Rice distribution $f_{A_s} = f(A_s \mid h\bar{A}_s, \sigma)$. Here $\bar{A}_s$ is the mean amplitude of the pure LoRa signal parts, $\sigma^2 = \frac{N_0}{2}$ and N_0 is the signal-sided noise power spectral density. Assuming equal probability for each LoRa symbol under our design, the average symbol confidence P_c can be expressed as

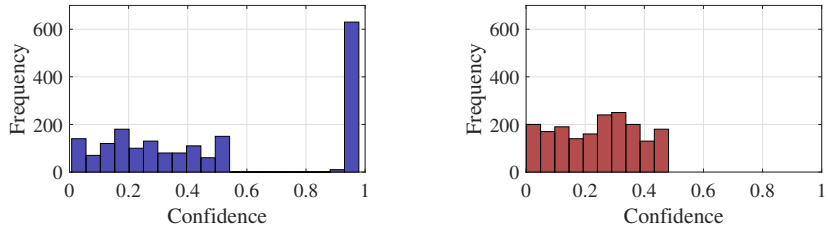
$$P_c = 1 - P'_{e|s} = 1 - \int_0^\infty \left[1 - F_\rho(x)\right] f_{A_s}(x)dx. \tag{13}$$

The symbol confidence P_c is jointly decided by noise power N_0 and symbol energy (which can be combined into SNR), as well as $\text{CTI}_{\text{ratio}}$ that is calculated in SLoRa+'s symbol recovery stage.

5.2 Spectrum-Based Symbol Confidence Correction

Section 5.1 theoretically analyzes the symbol confidence in AWGN channels. However, we find that in practice, the symbol confidence estimated from the theoretical model is not always correct. Specifically, to evaluate the limitations of our theoretical symbol confidence estimation, we conduct benchmark experiments with demodulating 200 LoRa packets in real-world scenarios and examine the confidence of incorrectly demodulated symbols. As shown in Figure 10(a), a large portion of incorrect symbols have surprisingly high confidence (i.e., $0.8 \sim 1$).

To address this issue, we propose a symbol confidence correction method that leverages the distinctive spectrum properties of LoRa to compensate for the diverse and unpredictable nature of real-world signal transmission. Specifically, we utilize a logistic binary classifier to categorize the **Single Frequency Factor Coefficient** (**SFFC**) features in the recovered symbol. The logistic classifier uses the sigmoid function to map the SFFC into probability values. The probability output



(a) Without correction (b) With correction

Fig. 10. Symbol confidence correction. (a) and (b) show the confidence distribution for incorrectly demodulated symbols without and with the spectrum-based confidence correction, respectively.

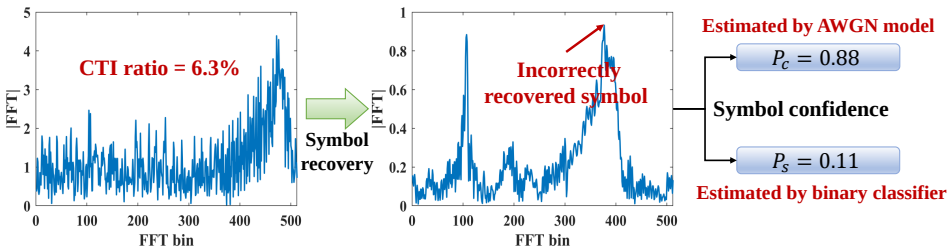


Fig. 11. Confidence estimation example for an incorrectly demodulated symbol.

from this classifier serves as a new metric to quantify whether the symbol is correctly demodulated.

$$P_s = \mathcal{F}(\text{SFFC}), \tag{14}$$

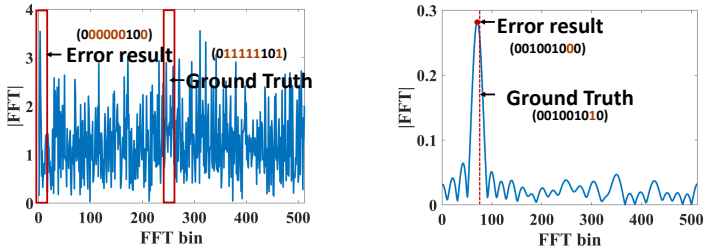
where P_s indicates the likelihood of accurate demodulation. $\mathcal{F}(\cdot)$ represents the function of logistic binary classifier. SFFC is calculated based on Equation (8), which examines the spectrum characteristics of LoRa and further demonstrates the reliability of the LoRa symbol. During the symbol recovery stage, SFFC examination is applied to these segments in the CTI set to reduce misclassification probability and achieve accurate CTI identification.

Figure 11 shows an example of symbol confidence estimation for an incorrect LoRa symbol after the symbol recovery module. The AWGN confidence model estimates a confidence (P_c) of 0.88. This is because even though the CTI duration ratio is only 6.3% in this symbol, the LoRa signal also suffers from background noises in a complex wireless channel. However, since the recovered symbol lacks a significant peak in the spectrum domain, it lacks the standard LoRa signal features. It is thus given low confidence (P_s = 0.11) via this spectrum-based classifier. Note that relying solely on the SFFC classification probability from Equation (14) to represent the demodulation confidence can also be inaccurate. As shown in Figure 12(b), after filtering out CTI-damaged LoRa signals, even though partially clean signals exhibit LoRa spectrum characteristics, it still cannot be demodulated correctly. In such cases, Equation (14) may assign an erroneously high confidence.

Therefore, we integrate our theoretical symbol confidence model with LoRa signal features to introduce the following symbol confidence estimation.

$$\hat{P}_c = \min\{P_c, P_s\}, \tag{15}$$

where P_c denotes the original theoretical confidence in Equation (13) and P_s represents the SFFC classification probability in Equation (14). The adjusted confidence $\hat{P}_c$ ensures a more accurate and reliable estimation of symbol confidence. Figure 10(b) shows the effectiveness of the spectrum-based



(a) Without a dominating peak (b) With a dominating peak

Fig. 12. Two patterns of FFT spectrum for incorrectly demodulated symbols. (a) Pattern 1: The spectrum does not have a dominating peak, which could lead to significant errors for all bits; (b) Pattern 2: The spectrum has a primary peak but has a slight deviation from the ground truth, indicating minor errors in several least significant bits.

correction. For symbols that are incorrectly demodulated and assigned high confidence, SLoRa+ reassigns appropriate lower confidence, i.e., smaller than 0.5, ensuring that confidence accurately reflects the communication reliability.

5.3 Accurate Bit Confidence Estimation

The above analysis assumes that all the bits within a LoRa symbol have the same reliability, indicating that the incorrect demodulation of a symbol would result in errors across all the bits. However, our real-world testing proves that some bits are inherently more reliable, which could be utilized to further improve the soft decoding performance of SLoRa+. Speficially, we identify two error patterns in symbols that have been incorrectly demodulated.

Pattern 1 (Figure 12(a)) is characterized by the FFT spectrum that lacks a dominating peak. It occurs when the LoRa signal is overwhelmed by noise or CTI. In this case, the recovered symbol could suffer from significant errors, so SLoRa+ treats all the bits equally.

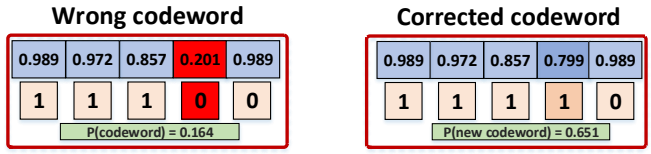
Pattern 2 (Figure 12(b)) is characterized by an FFT spectrum with a dominating peak, slightly misaligned from the true peak position, i.e., by approximately 4 FFT bins based on our observation. Adjacent frequency bins also exhibit high amplitudes. Such a pattern suggests minor errors in several least significant bits, while more significant bits are reliable. For example, as shown in Figure 12(b), the ground truth is 001001010 but SLoRa+ decodes it as 001001000, and only the second-to-last bit is incorrect. In our statistical experiments, pattern 2 appears in a larger proportion of all incorrectly demodulated symbols than pattern 1 (i.e., 75%).

For symbols with pattern 2, we propose to assign penalty weights to adjust the confidence of each bit within a symbol to achieve bit-level confidence estimation. This penalty specifically targets several least significant bits within a symbol because our analysis indicates they are more prone to errors. Specifically, the symbol confidence $\hat{P}_c$ is calculated based on Equation (15). Instead of giving each bit within the symbol the same confidence $\hat{P}_c$ [64]. We multiply the confidence for several least significant bits with a penalty weight α,

$$P_{\text{lsb}} = \hat{P}_c \times \alpha. \tag{16}$$

Our experiments empirically show that the three least significant bits of a symbol are particularly susceptible to errors. The penalty α is calculated using the amplitudes in the FFT spectrum,

$$\alpha = \frac{A_{f_d}}{\sum_{i=f_d-3}^{f_d+3} A_i}, \tag{17}$$



(a) The wrong codeword (11100) with its bit confidence

(b) The corrected codeword (11110) with its bit confidence

Fig. 13. One example of soft decoding.

where A_i signifies the amplitude of ith bin in the FFT spectrum. f_d is the dominating FFT bin. The penalty $\alpha < 1$ means the lower confidence of these least significant bits.

By adjusting the confidence of the three least significant bits based on the spectrum properties of the symbol, we provide a more accurate confidence assessment for each bit. With the estimated confidence for each LoRa bit, SLoRa+ further performs de-interleaving to get the code bit confidence within a codeword. Due to illustration purposes, these details are omitted. Figure 13(a) shows an example of the bit confidence within a wrong codeword with the coding rate 4/5. The red bit is erroneous due to CTI and exhibits lower confidence than other bits.

5.4 Soft Decoding

SLoRa+ adopts soft decoding [38] via the maximum likelihood searching for additional error correction capability. Denoting the codeword as R and the length of the codeword as n. The ith bit within R is represented as b_i. SLoRa+ examines all potential patterns within the codeword space C to compute the decoding result, selecting a codeword C_j with the highest probability.

$$P\big(R \mid C_j\big) = \text{argmax}_j \prod_{i=1}^{n} P\big(b_i = C_{ij} \mid C_{ij}\big)\,. \tag{18}$$

To reduce the search time, SLoRa+ further adopts a heuristic search. When a LoRa symbol suffers from more CTI, its bits are likely to be incorrect, which could be detected through the low confidence estimated. As a result, SLoRa+ initiates with low-confidence bits, iteratively flipping these bits to generate codewords in the codeword space, and outputs the codeword with the highest probability as the decoding result. Figure 13(b) demonstrates the result of soft decoding for Figure 13(a). SLoRa+ selects a codeword with the highest confidence and successfully corrects the corrupted symbol. The error correction capability of SLoRa+ is associated with the **coding rate** (**CR**). While traditional Hamming decoding fails to correct errors when CR = 4/5 and 4/6, SLoRa+ can correct one-bit errors when CR = 4/5, and two-bit errors when CR = 4/6. However, more bit errors might lead to decoding degradation in SLoRa+.

6 Theoretical Analysis

6.1 Accuracy Analysis

To assess the accuracy of symbol recovery and confidence estimation via theoretical derivation (Equation (13)), we conduct a simulation and transmit 10,000 LoRa symbols randomly within the AWGN channel under various CTI ratios and the same SNR (5 dB). Figure 14 illustrates that the calculated symbol recovery confidence values from SLoRa+ align closely with successful symbol demodulation ratios derived from the simulation. Additionally, the resilience of symbol recovery decreases as the proportion of the CTI increases. In addition, the increase of SF makes LoRa symbol a higher chance of successful recovery, because the larger SF, the longer the duration, the stronger the anti-interference ability, so as to provide better resilience.

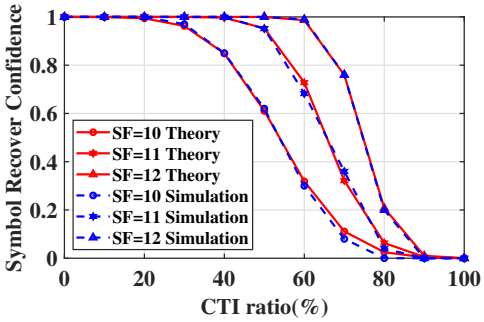


Fig. 14. Symbol recovery confidence from theoretical derivation and simulation under different CTI degrees.

6.2 Cost Analysis

Time complexity. SLoRa+ consists of two main modules: symbol recovery and soft decoding. We consider a target chirp signal that contains N samples. The complexity of symbol recovery comes from amplitude analysis and frequency analysis. The amplitude analysis involves a window-sliding algorithm to detect CTI, which has a complexity of approximately $O(N)$. The frequency analysis involves FFT-based processing of the LoRa chirp, with a complexity of $O(NlogN)$ The complexity of soft decoding is from symbol confidence estimation and maximum likelihood searching. Symbol confidence estimation is based on theoretical calculation and spectrum correction based on FFT processing, which has an approximate complexity of $O(NlogN)$. Soft decoding involves a maximum likelihood search. Due to the LoRa coding rules, it does not exceed 16 codeword combinations in the decoder [5], resulting in a decoding complexity of $O(16)$, which can be ignored because in most practical scenarios $N \gg 16$. Combining the complexities of both modules, the overall computational complexity of SLoRa+ can be expressed as $O(N + 2NlogN)$.

Space complexity. SLoRa+ is required to store the amplitude change and FFT results. Given that a LoRa symbol has N samples and that calculating amplitude and FFT operations do not occur concurrently, we choose the maximum storage space capable of accommodating both operations, equating to a $1 \times N$ double array.

In contrast, the time complexity of the state-of-the-art method PSR [47] scheme is $O\left(N^2 \times \log N\right)$. It uses a **short-time Fourier transform** (**STFT**), which requires traversing each signal window in the time domain and performing multiple FFTs. It also requires performing multiple STFTs with different window sizes for each LoRa symbol for fine-grained CTI detection. For the space complexity, it needs the storage space of approximately N^2 double matrices. For the standard LoRa, the demodulation cost is only $O(N \times \log N)$. And it needs a $1 \times N$ double array for FFT result storage. From the aforementioned discussion, SLoRa+ offers low time complexity and minimal storage space requirements, making it suitable for low-cost LoRa devices.

7 Evaluation

This section introduces the comprehensive evaluation of SLoRa+ through real-world testbeds, as shown in Figure 15. To evaluate the performance of SLoRa+, we test it under typical wireless interference in the 2.4 GHz band where SX1280 LoRa nodes operate. As Wi-Fi is the dominating interference in the 2.4 GHz, we first focus on the CTI due to Wi-Fi traffic and evaluate system performance and the impact of different parameter settings on SLoRa+ (Section 7.1 and Section 7.2). Considering the diversity of CTI sources and the frequency band where LoRa nodes operate, we also analyze the general performance of SLoRa+ under different CTI types (e.g., ZigBee and Bluetooth) and other LoRa operating frequency bands like 915 MHz and 433 MHz in Section 7.3.

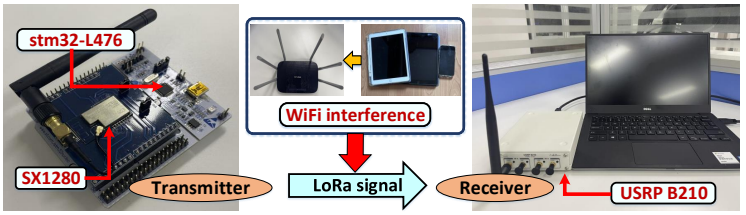


Fig. 15. Hardware setup.

Experiment setup. We use the commodity LoRa node as the transmitter. The LoRa node is composed of a STM32-L476 controller embedded with a Semtech SX1280 module [5]. The LoRa node transmits signals at the 2.4 GHz frequency band and the transmission power is set to 12.5 dBm. We employ a USRP B210 SDR [2] connected to a DELL XPS13 as the receiver. The GNU Radio software [1] is used to collect the wireless signal with a sampling rate of 500 kHz. We use MATLAB to process the raw physical samples and implement SLoRa+.

Comparison with existing works. Both the symbol recovery and soft decoding contribute to the effectiveness of SLoRa+. To demonstrate the contribution of each component, we conduct experiments to compare SLoRa+ with existing methods in terms of demodulation and decoding performance, respectively.

(1) **Evaluation on demodulation accuracy.** We compare SLoRa+ to the state-of-the-art method PSR [47] and the standard LoRa demodulation procedure which does not account for CTI (as described in Section 2.1). PSR attempts to reduce CTI impact by analyzing the signal feature and recovering the symbol at the physical layer.

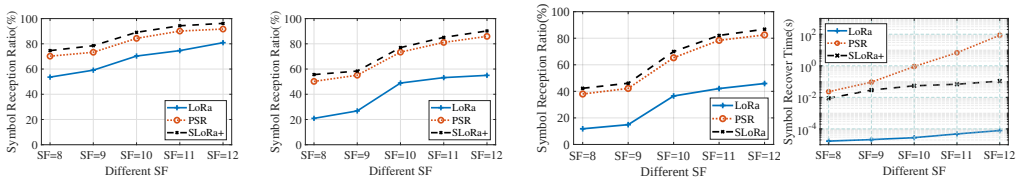
(2) **Evaluation on decoding accuracy.** We compare SLoRa+ with four other approaches: (1) Standard LoRa, which uses the typical LoRa demodulation and decoding techniques. (2) PLoRa, combining PSR's demodulation with traditional LoRa decoding. (3) DaRe [36], an application-layer coding scheme that ignores the physical layer signal. (4) SLoRa [64], SLoRa+'s conference version, which does not include the Spectrum-based Symbol Confidence Correction and Accurate Bit Confidence Estimation.

7.1 Evaluation on Demodulation Accuracy

This section analyzes the symbol recovery performance of SLoRa+. We evaluate how SLoRa+ performs under different interference levels and varying parameters.

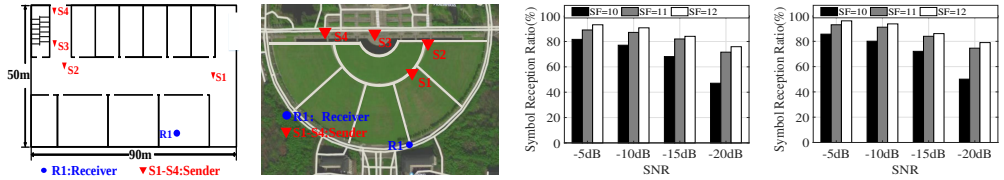
7.1.1 Performance with Different Interference Levels. We define interference levels (low, medium, and high) based on the interference density (i.e., the number of CTI packets generated by heterogeneous wireless devices per second). For Wi-Fi interference, we use commercial Wi-Fi devices (e.g., smartphones, iPads, and laptops) to generate Wi-Fi traffic at different volumes. Specifically, we connect 1, 3, and 5 devices to the Wi-Fi AP to generate low, medium, and high levels of interference, respectively. Low, medium, and high interference levels correspond to an average of 330, 1600, and 2700 Wi-Fi packets per second. For each spreading factor (SF) setting, we transmit more than 2,000 LoRa packets of 100 bytes in length with a packet transmission period of 1s. The LoRa node and the Wi-Fi device send packets simultaneously to generate CTI.

Figure 16 shows the **symbol reception ratio (SRR)** under different SF and interference levels. We can see that SLoRa+ ensures reliable symbol recovery. For example, with high interference and SF 10, the SRR for SLoRa+ is 69.9%, a substantial improvement over LoRa's 36.5%. A series of experiments show that SLoRa+ improves SRR across various interference conditions and SF settings.



(a) SRR (Low interference) (b) SRR (Medium interference) (c) SRR (High interference) (d) Symbol Recover Time

Fig. 16. Demodulation performance at the physical layer compared with the standard LoRa[5] and PSR[47]. (a)-(c) are the Symbol reception ratios under low, medium, and high interference. (d) Symbol Recover Time under different SFs.



(a) Indoor experiment scene (b) Outdoor experiment scene (c) SRR (Indoor) (d) SRR (Outdoor)

Fig. 17. (a) and (b) show the indoor and outdoor experimental scenarios, respectively. In both cases, LoRa senders are positioned from nearest to farthest at points S1-S4 under varying SNRs; (c) and (d) are the symbol recovery performance under different SNRs.

Furthermore, our design greatly accelerates computational speed while maintaining robust symbol recovery performance, a significant advantage over PSR. Figure 16(d) compares the symbol recovery time (SRT) of different methods. Under high interference level with SF 10, SLoRa+ achieves a similar SRR to PSR but runs 16.56× faster. As SF increases, the computational cost of PSR grows dramatically, whereas SLoRa+ remains highly efficient, requiring **less than 1%** of the computational resources of PSR. For example, when SF is equal to 12, PSR takes 86.28 seconds to restore a symbol, while SLoRa+ completes the task in just 0.11 seconds.

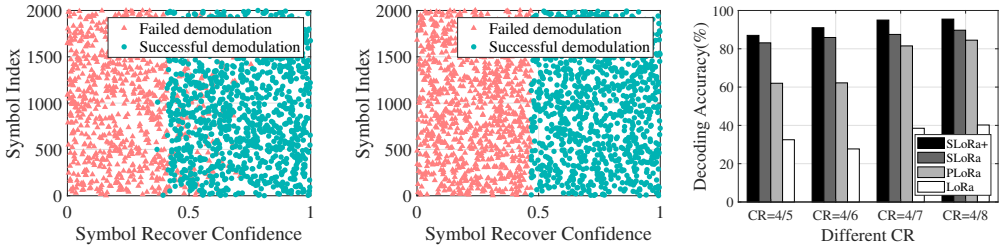
7.1.2 Performance under Different Parameters. SNR and SF are two key parameters that determine the accuracy of LoRa demodulation. SF determines both the chirp duration and noise resistance. SNR defines the ratio between the power of the received signal and the noise power (without including CTI power in our experiment). We evaluate SLoRa+'s robustness with varying SNR and SF in both indoor and outdoor environments. The indoor scenario is shown in Figure 17(a) where we conduct experiments in one typical 50 m × 90 m office building. Figure 17(b) shows the outdoor scenario, which is a 230 m × 500 m open university campus scene. We increase the distance between the transmitter and receiver to reduce the SNR from -5 dB to -20 dB with a step of 5 dB. SNR defines the ratio between the power of received signal and noise power, without including CTI power.

Impact of SNR. Figure 17(c) and Figure 17(d) show that the symbol reception ratio declines as SNR decreases. When the SNR is greater than -10 dB, SLoRa+ can successfully demodulate more than 80% of the symbols in all SFs. On the other hand, at lower SNRs (e.g., -20 dB), the symbol reception rate decreases due to noise. However, SLoRa+ can still recover more than 60% of the signal.

Impact of SF. As shown in Figure 17(c) and Figure 17(d), we observe that by comparing different SFs under a particular SNR, the SF of 11 and 12 always outperforms the SF of 10 for symbol reception. Specifically, in an indoor scenario with a SNR of -20 dB, SF = 10 yields a reception ratio of only 48.1%. In contrast, SF = 11 and SF = 12 result in much higher reception ratios of 72.5% and 80.1%, respectively. This highlights the resilience of higher SFs in challenging noise conditions as they can provide more clean LoRa samples for symbol recovery.

Table 1. Confusion Matrix of Logistic Classifier

Confusion Matrix		Prediction	
		Demodulate correctly	Demodulate wrongly
Ground Truth	Demodulate correctly	99.30%	0.70%
	Demodulate incorrectly	5.10%	94.90%



(a) Without symbol confidence correction (b) With symbol confidence correction (c) Decoding accuracy under CRs

Fig. 18. (a) and (b) The symbol recovery confidence vs demodulation results without and with the spectrum-based symbol confidence correction; (c) Decoding accuracy under different coding rates.

Impact of environment. We evaluate the performance of SLoRa+ in two different real-world environments, i.e., indoor (Figure 17(d)) and outdoor (Figure 17(d)). We observe that in the indoor environment, SLoRa+ achieves lower symbol reception ratios compared with the outdoor scenario. This is because the CTI, such as 2.4 GHz Wi-Fi and Bluetooth, is usually more severe in the indoor environment.

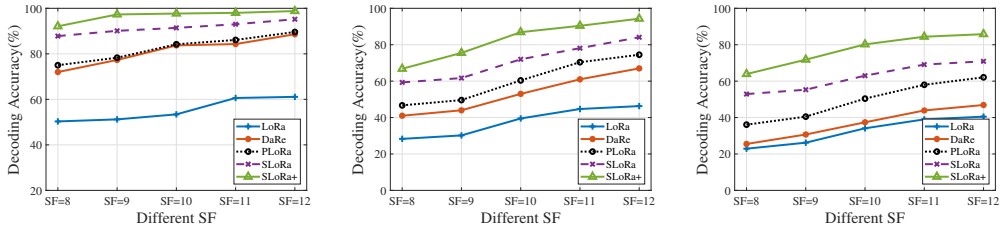
7.2 Evaluation on Decoding Accuracy

This section demonstrates the additional protection of soft decoding on top of symbol recovery.

7.2.1 Estimated Confidence for Recovered Symbols. In this section, we first prove the performance of the logistic binary classifier mentioned in Section 5.2, and then verify the effectiveness of SLoRa+'s symbolic confidence estimation algorithm.

Performance of logistic binary classifier. We first evaluate the performance of the proposed logistic binary classifier as demonstrated in Section 5.2. We extract 10,000 symbols from the data packets collected under different SNR and different interference levels as a dataset. The dataset is divided into training and test sets. 80% of the symbols are used for model training, while the test set includes the remaining 20%. We calculate the SFFC of each symbol as a binary classifier input feature for model training and testing. The model output is used as the confidence of symbol demodulation. Table 1 shows the confusion matrix for the logistic binary classifier. The True Positive ratio and True Negative ratio are 99.3% and 94.9%, respectively, demonstrating the accurate prediction of the logistic binary classifier. A False Positive ratio of 5.1% is caused by a lack of pure LoRa samples.

Performance of spectrum-based symbol confidence correction. Now we evaluate the effectiveness of our proposed confidence estimation for recovered symbols. Without loss of generality, we arbitrarily select 2000 corrupted symbols with SF = 9, 10, and 12. Then, we measure the CTI ratio for each symbol and utilize our symbol confidence model to calculate the symbol recovery confidence. Figure 18 shows the demodulation outcomes with symbol confidence correction. Figure 18(a) computes symbol confidence without applying symbol confidence correction. We observe that symbols with higher confidence are more likely to be successfully demodulated. However, some symbols, even with high confidence, can fail to be demodulated correctly. Our designed



(a) Decoding accuracy (Low CTI) (b) Decoding accuracy (Medium CTI) (c) Decoding accuracy (High CTI)

Fig. 19. Decoding performance compared with standard LoRa [5], PSR [47], DaRe [36], and SLoRa [64]. (a)–(c) are the decoding accuracy under low, medium, and high interference, respectively.

spectrum-based symbol confidence correction can further mitigate incorrect demodulation, thus improving the confidence estimation accuracy, as shown in Figure 18(b).

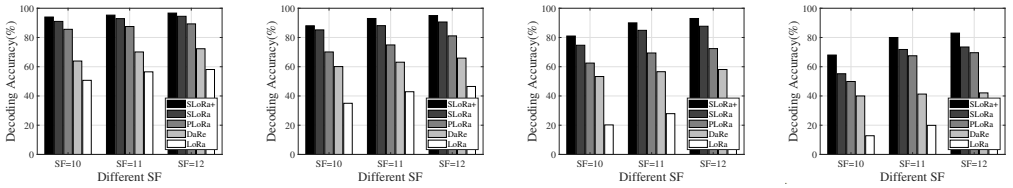
7.2.2 Performance with Different Interference Levels. Figure 19 illustrates the decoding performance of different interference levels and SF parameter configurations with a **coding rate** (**CR**) of 4/5. Without loss of generality, the experimental scenarios and parameter configuration remain the same as the experiment in the evaluation of demodulation accuracy, e.g., the SNR stands at -5dB, and the LoRa packet length is fixed at 100 bytes.

A comparison among SLoRa+, PLoRa, DaRe, and LoRa reveals that SLoRa+ is more suitable for CTI protection. For example, LoRa's decoding accuracy is merely 52.8% when SF = 10 under low interference. Employing PLoRa improves this decoding accuracy to 86.5%. The performance gain comes from the improved symbol reception ratio. DaRe's performance is similar to PLoRa's under low interference levels. However, our system SLoRa+ takes it a step further by integrating a soft decoding algorithm, pushing decoding accuracy up to 98.3%. With the increase of interference, LoRa suffers from more packet corruption, and the ability of DaRe and PLoRa to recover packets gradually decreases. But our design still maintains relatively stable data recovery performance. As shown in Figure 19(c), under high Wi-Fi interference level and SF 10, LoRa can only successfully decode 37.1% of the data. DaRe marginally enhances decoding accuracy by approximately 4%, and PLoRa decodes over 50% of data. SLoRa+ further augments the decoding accuracy to 80.2%.

Comparison. By comparing SLoRa and SLoRa+, we clearly show the effectiveness of spectrum-based symbol confidence correction and accurate bit confidence estimation, especially in the scenario with severe interference. Specifically, as shown in Figure 19(c), under SF = 10, SLoRa+ achieves a decoding accuracy of 80.2% while the accuracy of SLoRa is 62.4%, indicating **an improvement of 17.8%**. Moreover, the performance of SLoRa+ is **1.6**× that of PLoRa (i.e., 50.4%). This improvement primarily owes to a more precise confidence estimation of each LoRa symbol and a more dedicated mapping of symbol-to-bit confidence.

7.2.3 Performance with Different Parameters. To examine our system more thoroughly, we also conduct experiments to measure the decoding accuracy of SLoRa+, SLoRa, PLoRa, DaRe, and LoRa under varying SNR and SF conditions while maintaining the same CTI level. Given that interference levels or noise primarily influence the performance of demodulation or decoding, and the indoor environment suffers more serious interference, we only choose the indoor scene in Figure 17(a) for decoding performance evaluation in this section.

According to Figure 20, larger SF leads to higher decoding accuracy on all SNR levels. This can be attributed to the longer duration provided by larger SFs that enhance resilience. Notably, while



(a) Decoding accuracy(-5 dB) (b) Decoding accuracy(-10 dB) (c) Decoding accuracy(-15 dB) (d) Decoding accuracy(-20 dB)

Fig. 20. Decoding accuracy under different SNR and SF.

extremely low SNR causes a substantial challenge to LoRa decoding, SLoRa+ still demonstrates high decoding accuracy with large SFs. For example, when SNR is at -20 dB, the decoding accuracy for SF 11 in LoRa merely reaches 19%, whereas over 80% bytes are successfully decoded at SF $\geq$11 in SLoRa+. On the other hand, the enhanced design in SLoRa+ delivers robust performance under diverse SNRs, yielding a 2.7% $\sim$ 11% improvement in decoding accuracy for SLoRa.

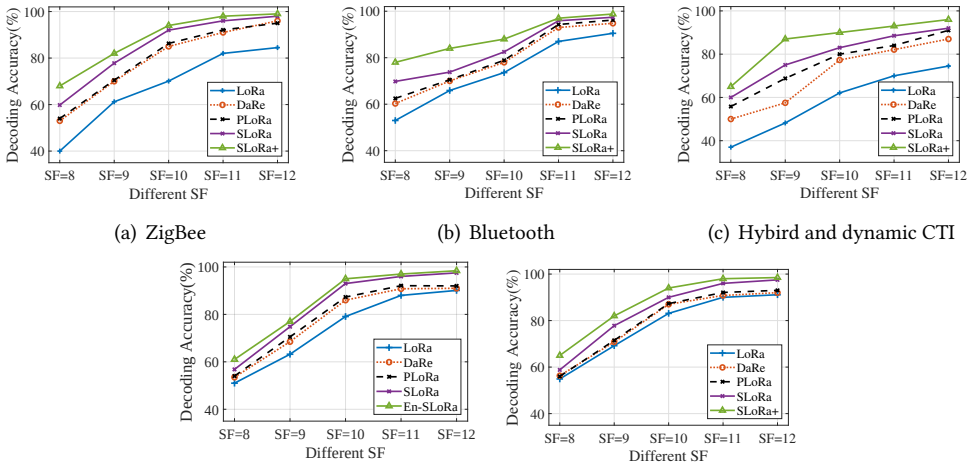
7.2.4 Performance with Different CR. This section evaluates the SLoRa+'s performance under different coding rates. We set the coding rates as 4/5, 4/6, 4/7, and 4/8, respectively, with other parameters being SF 10 and SNR = -10 dB. As shown in Figure 18(c), SLoRa+ consistently performs well regardless of the coding rate. For example, when CR is set to 4/6, the decoding accuracy of SLoRa+ is 1.47 times that of PLoRa, reaching 91.0%, while the decoding accuracy of LoRa is less than 28%. The performance gains are greater when CR is 4/5 or 4/6 because the standard LoRa utilizes Hamming decoding, which merely detects errors but falls short in correcting them under CR = 4/5 or 4/6. When CR is 4/7 and 4/8, it can correct one-bit errors but is unable to manage multi-bit errors, which is not enough to eliminate the adverse effects of CTI. SLoRa+ decodes based on each bit's confidence, enabling error correction even if CR is 4/5. The comparison between SLoRa and SLoRa+ shows the effectiveness of the enhanced designs in SLoRa+. For example, when CR equals 4/7, SLoRa+ enhances SLoRa's decoding accuracy from 87.5% to 94.5%.

It's worth noting that in these experiments, SLoRa+ does not guarantee a 100% decoding accuracy. This is because SLoRa+ strictly adheres to the standard LoRa packet structure and coding algorithm. Although standard LoRa leverages interleaving to combat burst interference, this technique only applies interleaving for bits within a single block (in our case, a block is m $\times$ SF, where m represents the codeword length). As a result, it is possible that large errors exist within one codeword, leading to decoding inaccuracies. If interleaving is applied at the packet level, SLoRa+ would offer more resilience via its soft decoding approach.

7.3 General Protection

SLoRa+ does not need to have any prior knowledge of interference in advance. Therefore, our system adopts a general design, which can effectively recover LoRa symbols under various interference and different frequency bands.

7.3.1 Performance with Different CTIs. To verify the universality of SLoRa+ under different CTIs, we test its performance in the presence of ZigBee and Bluetooth interference. Both ZigBee and Bluetooth operate in the 2.4 GHz ISM band and can negatively affect LoRa communication. Specifically, we use a CC2530 ZigBee node and a CC2640 Bluetooth node to generate CTI during the LoRa signal transmission period. The packet transmission rates of Zigbee and Bluetooth are 20 packets per second and 100 packets per second, respectively. The transmission power is set to 5 dBm for the ZigBee node and 0 dBm for the Bluetooth node, and the distance between the interference source and the LoRa receiver is maintained at 2.5 meters.



(a) ZigBee (b) Bluetooth (c) Hybird and dynamic CTI

(d) 915 MHz (e) 433 MHz

Fig. 21. General protection of SLoRa+: (a) and (b) Decoding accuracy under different CTIs such as ZigBee and Bluetooth; (c) Decoding accuracy under hybrid and dynamic CTIs; (d) and (e) Decoding accuracy under other LoRa working frequency bands, such as 915 MHz and 433 MHz, which could be corrupted by IEEE 802.11.ah, telemetry transmitters, other LPWAN technologies such as Weightless and ZETA.

Figure 21(a) and 21(b) show the decoding accuracy of SLoRa+ under these two types of interference. SLoRa+ maintains robust performance under diverse SF configurations, demonstrating its ability to adapt to diverse CTI scenarios. As shown in Figure 21(a), SLoRa+ achieves high decoding accuracy under ZigBee interference, particularly at higher SF values. However, when the SF is small, e.g., SF8, the performance degrades. This is due to the long packet duration of ZigBee, which prevents SLoRa+ from finding enough clean LoRa samples for symbol recovery. Figure 21(b) shows the performance of SLoRa+ under Bluetooth interference. We can observe that SLoRa+ achieves higher decoding accuracy compared with ZigBee interference in low-SF configurations. This is because the shorter packet duration of Bluetooth allows SLoRa+ to locate clean LoRa samples more effectively. Compared with Wi-Fi interference, both ZigBee and Bluetooth have less impact on LoRa communication due to their lower power levels. In real-world IoT settings, where multiple wireless technologies coexist and produce diverse CTIs, we expect SLoRa+ to provide considerable resilience, although its performance may vary with the strength and nature of the interference.

7.3.2 Performance with Dynamic and Hybrid CTI. To evaluate the effectiveness of our method in handling various interference sources in complex scenarios, we experiment with one LoRa node and three different interference sources. We deploy a Bluetooth node and a ZigBee node in the indoor scenario shown in Figure 17(a), and use a smartphone to connect to the Wi-Fi AP. The interference sources transmit packets at consistent rates, with Wi-Fi, ZigBee, and Bluetooth sending approximately 330, 20, and 100 packets per second, respectively.

In this experiment, the LoRa node transmits signals to the receiver while the interference sources generate different types of CTI. These interference sources move randomly in the indoor environment at an average speed of about 1 m/s. Figure 21(c) illustrates the performance of SLoRa+ in complex scenarios where multiple dynamic interference sources coexist. The results show that SLoRa+ consistently provides robust protection for LoRa communication, even in dynamic environments with multiple CTI sources. This is because SLoRa+ operates without relying on any prior knowledge of the CTI, such as its type or motion state.

7.3.3 Performance with Different Frequency Bands. We also test the general feasibility of SLoRa+ in different LoRa operating bands (i.e., 915 MHz and 433 MHz) using LoRa SX1276 nodes. It is worth noting that in the 915 MHz band, LoRa may suffer from interference from IEEE 802.11.ah technologies, while in the 433 MHz band, potential interference sources range from phones, wireless headsets, and telemetry transmitters to LPWAN technologies such as Weightless and ZETA. Given the wide variety of applications in the sub-1 GHz ISM bands, mutual interference between various wireless protocols is quite significant. Figure 21(d) and Figure 21(e) show the decoding performance of different methods. Similar to 2.4 GHz, LoRa also suffers from CTI in other working bands. However, since the power of CTI below sub-1GHz is lower than that of 2.4 GHz CTI such as Wi-Fi, the severity of interference is less than that of the 2.4 GHz band. Through symbol recovery and soft decoding, SLoRa+ enhances reliability against interference and maintains high decoding accuracy under CTI.

8 Discussion

Severe interference challenges. Our design effectively addresses the CTI issue through techniques like symbol recovery and soft decoding. However, the system performance degrades under extreme interference. When CTI damages more than 80% of LoRa symbols, our symbol recovery may not succeed, because in such extreme collision scenarios, the number of undistorted chips within a LoRa symbol becomes insufficient for successful recovery. To further enhance performance, SLoRa+ proposes soft decoding. The error correction capability of SLoRa+ is influenced by the coding rate (CR). While traditional Hamming decoding struggles with error correction at CR=4/5 and 4/6, SLoRa+ can correct one bit error at CR=4/5 and two bits at CR=4/6. However, more bit errors may make error correction more difficult, resulting in the degradation of SLoRa+'s decoding performance. In the future, we plan to leverage deep learning techniques to tackle this challenge.

LH-FHSS modulation. It is worth noting that Semtech introduced an extension to the LoRa physical layer called LR-FHSS (Long Range Frequency Hopping Spread Spectrum) [9, 10]. LR-FHSS splits a LoRa packet's payload into fragments and uses fast frequency hopping to transmit those fragments, thereby avoiding long channel occupancy and improving network capacity. This symbol-level frequency hopping requires careful coordination between the transmitter and receiver. Currently, only a limited number of LoRa devices support LR-FHSS (e.g., SX126x series[3], LR1121 [6]). In addition, LoRa devices also support frequency hopping at the packet level. A LoRa end device has eight channels of 125 kHz with 200 kHz spacing and it can choose to send packets on different channels [32]. We believe that combining SLoRa+ and frequency hopping function can further enhance the LoRa communication reliability under interference.

Deployment. SLoRa+ is a physical-layer software solution focused on solving the impact of CTI on LoRa communication. It is suitable for packet loss caused by strong interference in low SNR environments, which usually occurs in high-density areas where a large number of heterogeneous devices coexist. Without any hardware modification, SLoRa+ can be quickly deployed on the LoRa receiver and is compatible with existing LoRa transmitters.

9 Related Work

9.1 LoRa Communication

LoRa facilitates long-range wireless connections with low power consumption [7, 41]. In recent years, LoRa technology has made great progress in throughput [61], weak link reception [24, 28, 62], and scalability [73, 74], etc. To achieve high throughput, Hylink [61] proposes a parallel CSS technique (called p-CSS) that modulates multiple symbols during the chirp duration to speed up the bit rate on LoRa links. To improve LoRa communication over weak links, NELoRa [28] employs a mask-enabled deep neural network to enhance ultra-low SNR LoRa demodulation. MALoRa [24]

exploits the multi-antenna capability of LoRa gateways to boost weak packet reception. XCopy [62] coherently integrates the signal copies of the same data packet in multiple retransmissions to obtain a greater SNR gain. To improve the scalability of LoRa networks, XGate [73] introduces an auto-configured LoRa gateway capable of utilizing a single Rx chain to efficiently detect incoming packets across all available logical channels. In contrast, SLoRa+ focuses on a different and important problem in LoRa communication - CTI.

9.2 LoRa Communication under Interference

As it operates within the unlicensed ISM band, LoRa inevitably suffers from either homogeneous interference caused by concurrent transmissions of LoRa devices or heterogeneous interference across different wireless protocols [22, 45]. Wireless interference can easily corrupt LoRa packets, resulting in extended transmission delays and additional energy consumption.

Concurrent LoRa transmission. Recent research has been primarily focused on reliable LoRa communication under concurrent LoRa interference. Such studies can be broadly categorized into three main streams. The first category involves processing received physical signals to separate concurrent LoRa transmissions [12, 13, 15, 25, 29, 35, 44, 53, 57–59, 63, 68, 71, 72]. These techniques aim at identifying unique signal features, such as central frequency offset, frequency and amplitude pattern, or phase diversity associated with different LoRa devices. Hi2LoRa [12] expands the feature dimension by extracting the inherent channel state information of each LoRa transceiver pair and tracking the temporal pattern of the features, thereby enhancing the concurrent decoding performance. The second line of approaches [11, 17, 37, 49] focuses on designing dedicated encoding and decoding schemes to enhance network scalability. The third type of method [18, 65, 66] focuses on MAC layer scheduling and effectively avoids transmission conflicts by adjusting parameter configuration (e.g., SF, transmission power, channel, transmission time).

Heterogeneous interference (CTI). Unlike interference caused by concurrent LoRa transmissions, CTI originates from heterogeneous wireless technologies that are fundamentally different from LoRa [47]. This heterogeneity makes existing methods designed for addressing concurrent LoRa interference ineffective in handling CTI. To tackle this issue, the signal analysis techniques, i.e., PSR[47], propose partial symbol recovery via Short-Time Fourier Transform. Additionally, several fFEC techniques [36, 39] incorporate extra coding bits to correct errors in received LoRa packets. Different from the existing literature on CTI protection, SLoRa+ introduces a lightweight symbol recovery scheme followed by a confidence analysis of the recovered symbols. This information is subsequently leveraged via soft decoding to enhance reliability. In contrast, previous signal analysis techniques ignore specific CTI situations. They blindly treat all recovered symbols as of equal quality and also require intensive computational resources. As for existing encoding solutions, they ignore critical physical layer information that could be used to significantly improve reliability. It is worth noting that SLoRa+'s symbol recovery and confidence estimation directly rely on LoRa's unique signal characteristics under CTI and are therefore fundamentally different from previous soft decoding methods [49, 69, 75].

9.3 Cross-Technology Communication

Cross-Technology Communication (**CTC**) is a technique that enables direct data exchange between devices operating on different wireless technologies [14, 19–21, 26, 31, 33, 34, 46, 52, 60], without requiring hardware modifications. By leveraging CTC, existing methods enable channel coordination [31, 55, 56, 70], i.e., heterogeneous devices can exchange information about future channel usage to minimize CTI. However, implementing CTC-based transmission coordination in LoRa networks presents challenges. First, CTC coordination requires heterogeneous devices to be controllable at the software level, which adds implementation complexity. Second, due to LoRa's

long-range communication and the weak received signal, which falls below the noise level, CTC coordination may require introducing transmission delays for heterogeneous devices distributed over large areas. In contrast, SLoRa+ does not rely on controlling heterogeneous devices. It focuses on mitigating CTI at LoRa's physical layer to ensure reliable LoRa communication.

10 Conclusion

This article introduces SLoRa+, a lightweight and effective framework designed to enhance the resilience of LoRa networks against CTI. SLoRa+ achieves this by integrating symbol recovery with advanced soft decoding strategies. In the soft decoding module, two dedicated designs including spectrum-based symbol confidence correction and accurate bit confidence estimation, can ensure the confidence estimation accuracy and further enhance the robustness against CTI in LoRa communication. Comprehensive theoretical analysis and extensive testbed-based experiments verify the performance of SLoRa+ in providing CTI protection under various settings.

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